

# CS 486/686

# Heuristic Search

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Lecture 3

RN 3.5 · PM 3.6–3.7

Search ›

Uncertainty ›

Decisions ›

Learning

## Learning goals

- Motivate heuristic search
- Trace LCFS, GBFS, A\*
- Analyze: space, time, completeness, optimality
- Design and dominate admissible heuristics
- Prune the search space

# Which state is closer to the goal?

State A

|   |   |   |
|---|---|---|
| 5 | 3 |   |
| 8 | 7 | 6 |
| 2 | 4 | 1 |

State B

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 |   |
| 7 | 8 | 6 |

Goal

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 |   |

**B is just one move away.** Humans see this instantly — uninformed search does not.

# Uninformed vs heuristic

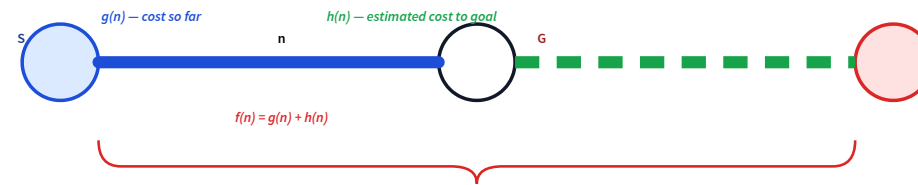
## Uninformed

- Treats every state the same
- No idea which is closer to goal

## Heuristic

- Estimates distance to goal
- Expands the most promising state

# Two quantities at every node $n$



Solid **blue**: known. Dashed **green**: estimated. **Red**: total estimate.

# What makes a good heuristic?



**Problem-  
specific**



**Non-negative**



$h(\text{goal}) = 0$



**Cheap to  
compute**

# Three algorithms, one knob

Frontier is a priority queue. The three algorithms differ in the priority key.

LCFS

$$g(n)$$

cheapest-path-so-far

GBFS

$$h(n)$$

looks-closest-to-goal

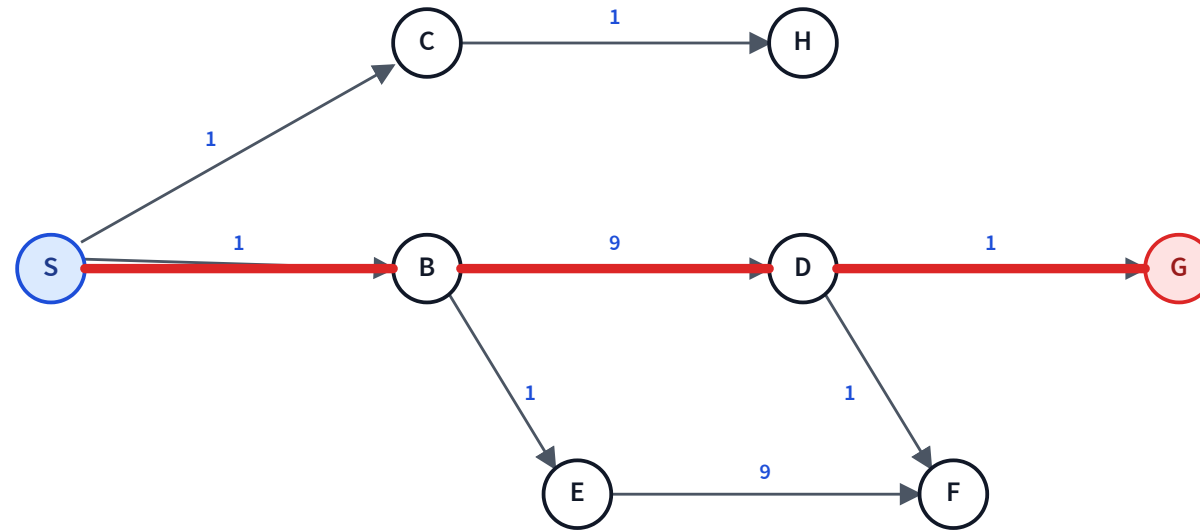
A\*

$$g(n) + h(n)$$

total estimated cost

# The trace graph

All four traces use this graph. Edge labels are **costs**. Note that F has two parents (E and D).



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Optimal path: S → B → D → G with total cost **11**.



## Lowest-cost-first search (LCFS)

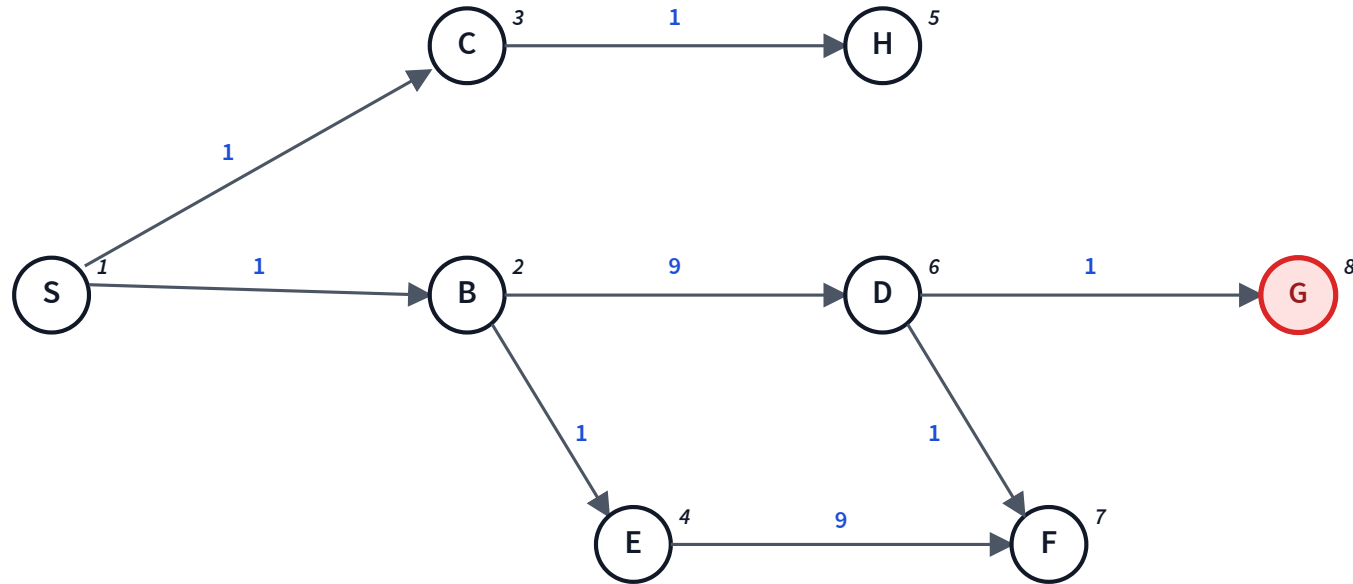
- Frontier = **priority queue**, key =  $g(n)$
- Pop the cheapest path so far

a.k.a. **Uniform-cost search (UCS)** in RN, or **Dijkstra's algorithm** in algorithms textbooks.

# Trace: LCFS

Path notation SBE: 2 = path  $S \rightarrow B \rightarrow E$  with cumulative cost 2. Ties broken alphabetically.

**Frontier:**



# LCFS — properties

|           |                                                                                    |
|-----------|------------------------------------------------------------------------------------|
| Space     | $O(b^d)$ <i>exponential</i>                                                        |
| Time      | $O(b^d)$ <i>exponential</i>                                                        |
| Complete? | <b>Yes</b> <i>finite branching, edge cost <math>\geq \varepsilon &gt; 0</math></i> |
| Optimal?  | <b>Yes</b> <i>under the same conditions</i>                                        |

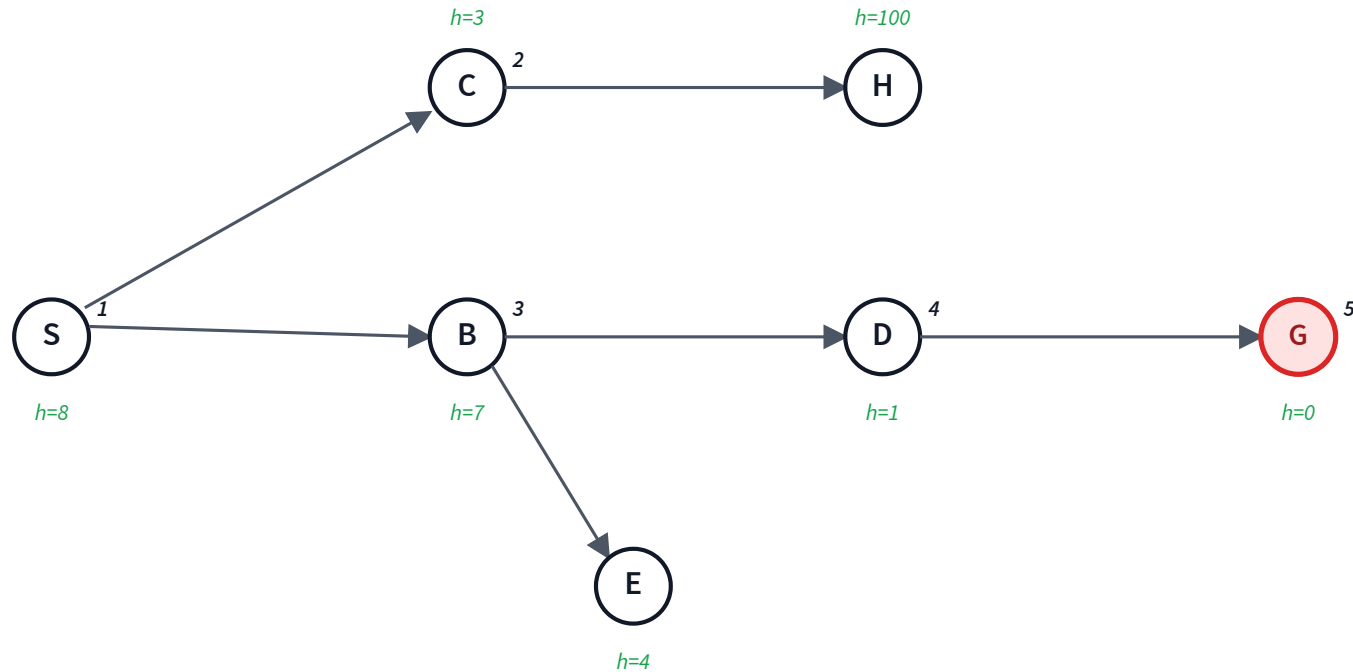
## Greedy best-first search (GBFS)

- Frontier = **priority queue**, key =  $h(n)$
- Pop the most promising path (lowest  $h$ )

# Trace: GBFS

Path notation SBD: 1 = path with  $h(D) = 1$ . Heuristic values shown in green.

Frontier:



# GBFS — properties

|           |                                                                 |
|-----------|-----------------------------------------------------------------|
| Space     | $O(b^m)$ <i>exponential</i>                                     |
| Time      | $O(b^m)$ <i>exponential</i>                                     |
| Complete? | <b>No</b> <i>can get stuck following a misleading heuristic</i> |
| Optimal?  | <b>No</b> <i>greedy choice may be wrong</i>                     |

# Intuition: what each algorithm prioritises

|                           | LCFS                                                              | GBFS                                                            |
|---------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------|
| Frontier sorted by        | $g(n)$ — cost so far                                              | $h(n)$ — estimated cost to goal                                 |
| Closest uninformed cousin | BFS with weights.<br>When every edge costs 1, LCFS = BFS exactly. | DFS guided by $h$ .<br>Chases one direction aggressively.       |
| Search shape              | A cost-wavefront that grows <i>outward</i> from the start.        | A narrow probe that dives toward whatever <i>looks</i> closest. |

# Intuition: what guarantees follow

|           | LCFS                                                             | GBFS                                                            |
|-----------|------------------------------------------------------------------|-----------------------------------------------------------------|
| Complete? | <b>Yes</b> — wavefront eventually reaches every reachable state. | <b>No</b> — low- $h$ cycles can trap it.                        |
| Optimal?  | <b>Yes</b> — first pop of $G$ has the smallest $g$ .             | <b>No</b> — $h$ alone can prefer a costly “looks-close” detour. |

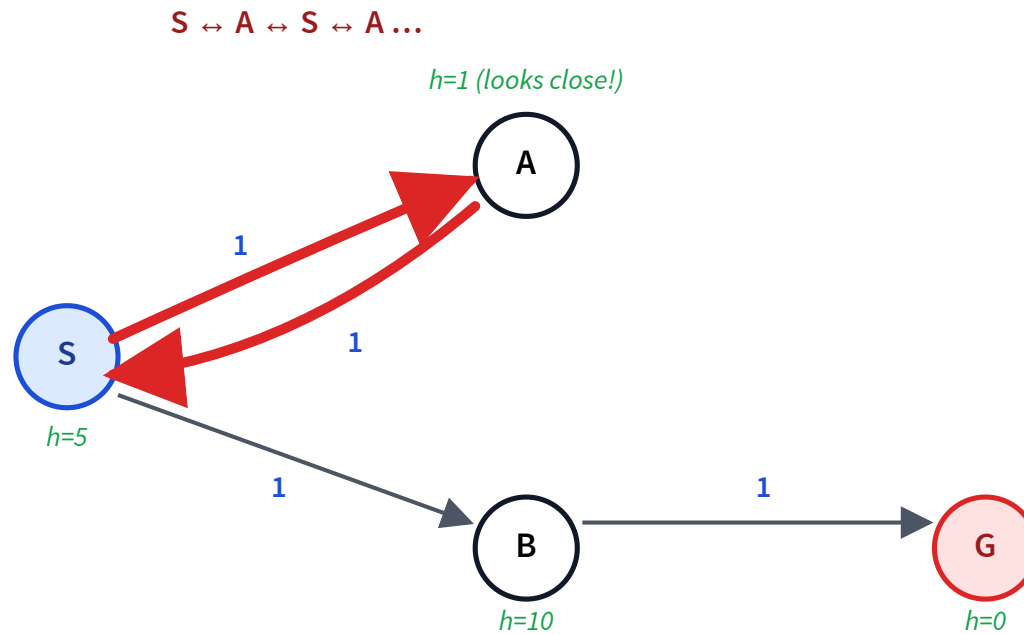
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Worst-case time / space: LCFS is  $O(b^d)$ ; GBFS is  $O(b^m)$ .



# GBFS — not complete

A low- $h$  cycle can trap GBFS forever — it never explores the path that actually reaches G.

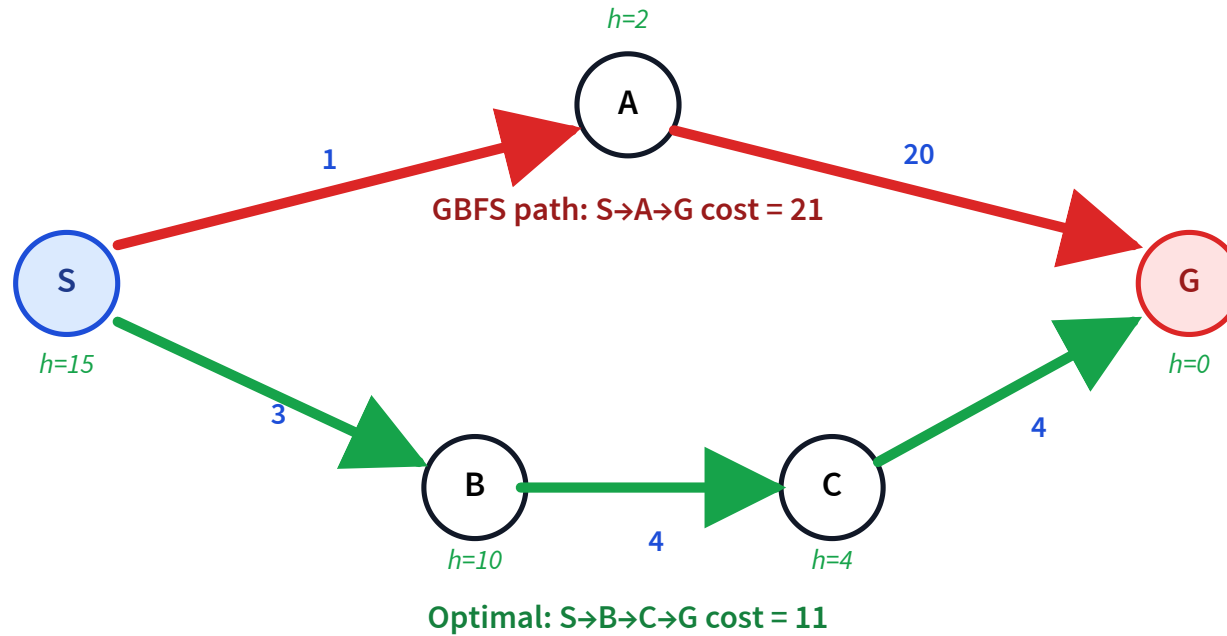


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Pop order: SA ( $h=1$ ), SAS ( $h=5$ ), SASA ( $h=1$ ), ... SB ( $h=10$ ) is always outranked — G never reached.

# GBFS — not optimal

Even when GBFS finds the goal, the path can be far from cheapest.



## A\* search

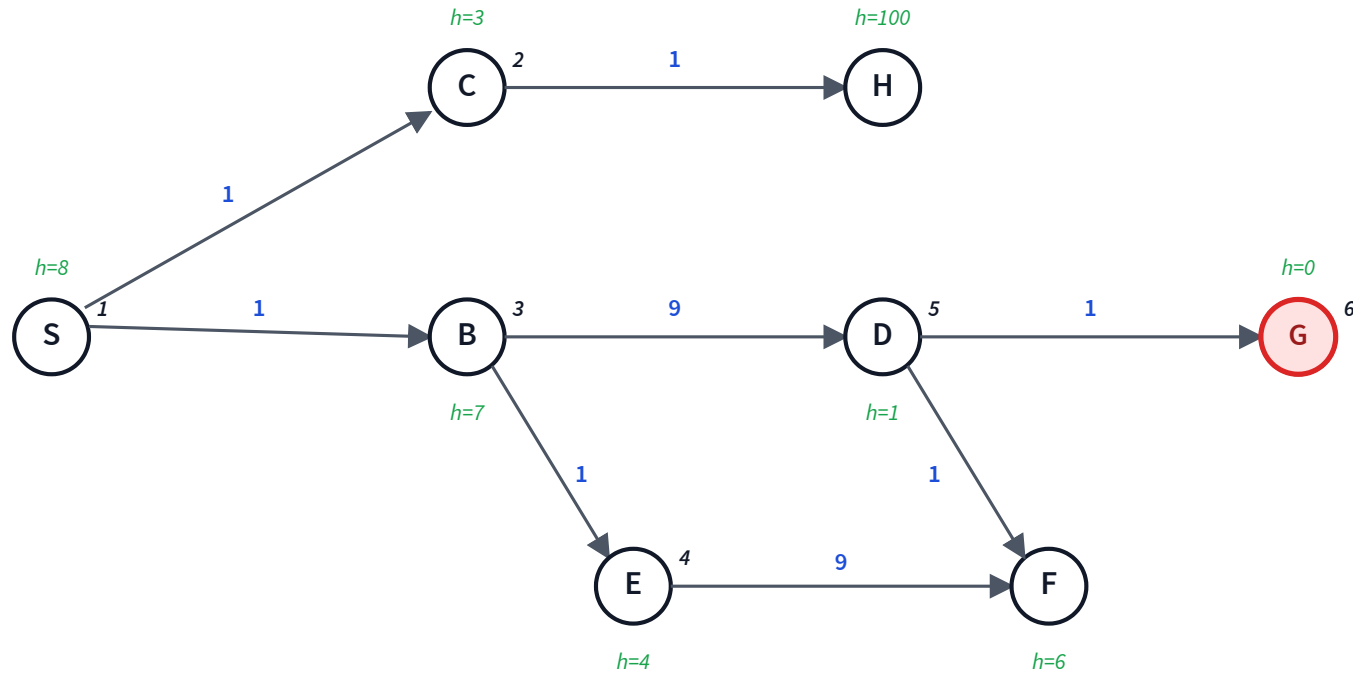
- Frontier = **priority queue**, key =  $f(n) = g(n) + h(n)$
- Pop the path with smallest total estimated cost

A\* combines LCFS's **cost-so-far** with GBFS's **goal-pull**.

# Trace: A\*

Path notation SBE: 6 = path with  $f = g + h = 6$ . **g** in blue, **h** in green.

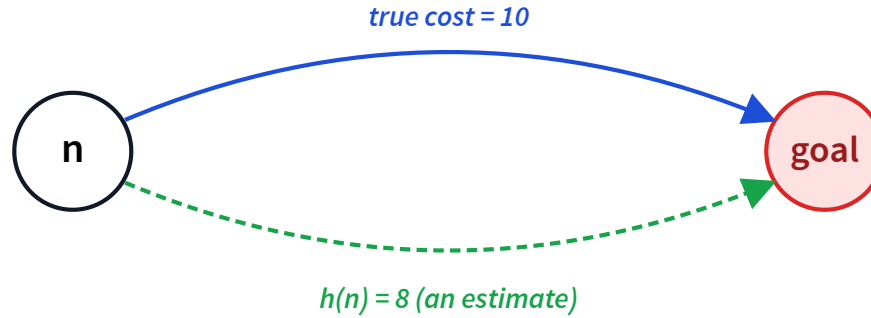
**Frontier:**



# A\* — properties

|           |                                                              |
|-----------|--------------------------------------------------------------|
| Space     | $O(b^m)$ <i>worst case; often much better</i>                |
| Time      | $O(b^m)$ <i>worst case; often much better</i>                |
| Complete? | <b>Yes</b> <i>under the usual finite-branching condition</i> |
| Optimal?  | <b>Yes</b> <i>if <math>h</math> is admissible</i>            |

# Admissible heuristic



**Admissible:**  $h(n) \leq$  true cost from  $n$  to any goal, for every  $n$ . *Never over-estimates.*

**Theorem.** If  $h$  is admissible,  $A^*$  returns an optimal path.

# A\* optimality — proof sketch

Assume A\* returns cost  $C^*$ , but the true optimum is  $C' < C^*$ .

1. Some node  $N$  on the optimal path is still on the frontier at termination.
2. Admissibility  $\Rightarrow f(N) = g(N) + h(N) \leq C'$ .
3. So  $f(N) \leq C' < C^*$  — A\* would have popped  $N$  first. **Contradiction.** ■

**$A^*$  is optimally efficient**

No other optimal algorithm using the same  $h$  expands fewer nodes.



# Manhattan distance

Sum of grid distances of each tile to its goal position.

| Initial  |   |   | Goal |   |   |   |   |   |
|----------|---|---|------|---|---|---|---|---|
| 5        | 3 |   | 1    | 2 | 3 |   |   |   |
| 8        | 7 | 6 | 4    | 5 | 6 |   |   |   |
| 2        | 4 | 1 | 7    | 8 |   |   |   |   |
| tile     | 1 | 2 | 3    | 4 | 5 | 6 | 7 | 8 |
| distance | 4 | 3 | 1    | 2 | 2 | 0 | 2 | 2 |

$$\text{Manhattan}(\text{start}) = 4 + 3 + 1 + 2 + 2 + 0 + 2 + 2 = 16$$

# Misplaced tile count

Number of tiles not yet in their goal position.

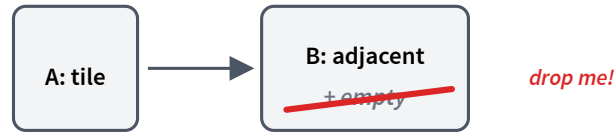
| Initial |   |   | Goal |   |   |
|---------|---|---|------|---|---|
| 5       | 3 |   | 1    | 2 | 3 |
| 8       | 7 | 6 | 4    | 5 | 6 |
| 2       | 4 | 1 | 7    | 8 |   |

Red = misplaced. Green = already at goal.

$$\text{Misplaced}(\text{start}) = 7$$

Both heuristics are **admissible**.

# Recipe: relax the problem



1. **Relax** the problem by dropping constraints.
2. **Solve** the relaxed problem (without search).
3. That cost is an **admissible heuristic** for the original.

An easier problem gives a lower bound — safe to use as an estimate.

# Quick check: heuristic from relaxation

**Q.** Drop the "B is empty" constraint: a tile can move from A to B if A and B are adjacent. Which heuristic does this give?

- A. Manhattan distance
- B. Misplaced tile count
- C. Another heuristic not listed

**A — Manhattan distance.** Without the "B must be empty" constraint, each tile slides freely along the grid to its goal — cost = number of grid steps = Manhattan distance.

# Dominating heuristics

**Definition.**  $h_2$  dominates  $h_1$  if  $h_2(n) \geq h_1(n)$  for all  $n$ , and  $h_2(n) > h_1(n)$  for some  $n$ .

**Theorem.** If  $h_2$  dominates  $h_1$ ,  $A^*$  with  $h_2$  expands no more nodes than  $A^*$  with  $h_1$ . Higher = better — as long as still admissible.

# Manhattan vs Misplaced

**Q.** On our example state, Manhattan = 16 and Misplaced = 7. Which heuristic dominates the other in general?

- A. Manhattan dominates Misplaced
- B. Misplaced dominates Manhattan
- C. Neither dominates the other

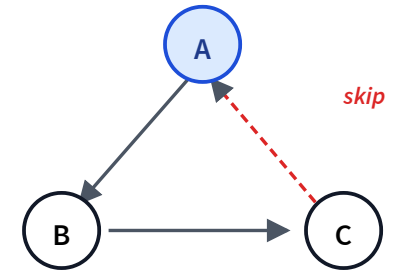
**A — Manhattan dominates.** Each misplaced tile contributes at least 1 to both heuristics; Manhattan adds the actual grid distance, so  $\text{Manhattan} \geq \text{Misplaced}$ , and strictly  $>$  whenever any tile is more than one step from its goal.

# Cycle pruning

Don't follow a path that revisits a node already on the current path.

When expanding path  $\langle n_0, \dots, n_k \rangle$ :

1. For each neighbour  $n$  of  $n_k$ :
2. If  $n$  already appears in  $\langle n_0, \dots, n_k \rangle \rightarrow$  **skip**.
3. Else add  $\langle n_0, \dots, n_k, n \rangle$  to the frontier.



Check is linear in path length. Saves DFS from infinite loops.

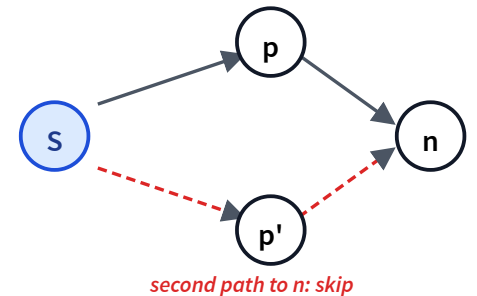
# Multi-path pruning

Don't expand a node we've already expanded (across *different* paths). Cycle pruning is a special case.

Maintain an explored set, initialized to  $\{ \}$ .

**Pop path  $\langle n_0, \dots, n_k \rangle$ . Then:**

1. If  $n_k \in \text{explored}$   $\rightarrow$  skip.
2. Add  $n_k$  to explored.
3. If  $n_k$  is the goal  $\rightarrow$  return the path.
4. Else push each neighbour-extension to the frontier.





## Caveats of multi-path pruning

- Repeats still pushed — skipped on pop
- Trades memory (explored set) for time
- May sacrifice optimality (see next slides)

# LCFS + multi-path pruning

Q. LCFS with multi-path pruning — can it discard the optimal solution?

A. Yes, it can

B. No, never

**B — No.** First pop of any node already has the smallest  $g$ ; later paths to it can only be costlier.  
Safe to prune.

# A\* + multi-path pruning

Q. A\* with admissible  $h$  + multi-path pruning — can it discard the optimal?

A. Yes, it can

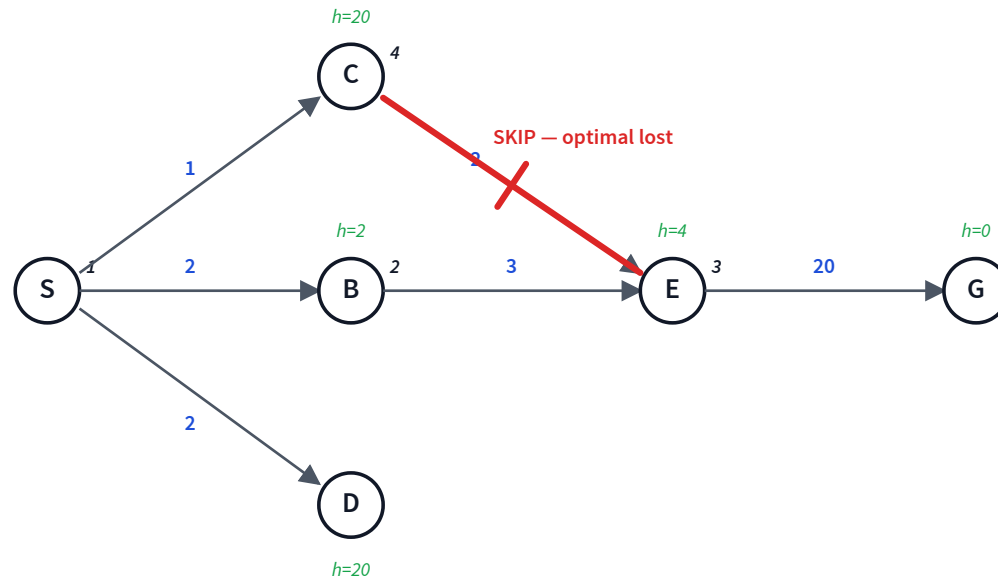
B. No, never

**A — Yes, it can.** A\* orders by  $f = g + h$ , so the first pop of  $n$  need not have the smallest  $g$ . A cheaper  $g(n)$  found later is then discarded. *Counterexample next.*

# A\* + multi-path: counterexample

Optimal path is S→C→E→G (cost 23). With multi-path pruning, A\* returns S→B→E→G (cost 25) instead.

**Frontier & Explored:**



# Consistent (monotone) heuristic

**Definition.**  $h$  is consistent if for every edge  $m \rightarrow n$ :

$$h(m) \leq \text{cost}(m, n) + h(n)$$

i.e., the heuristic decreases by no more than the edge cost as we move toward the goal.

**Theorem.** If  $h$  is consistent, A\* with multi-path pruning is **optimal**.

Most natural heuristics (Manhattan, straight-line distance, ...) are consistent.

# Where $A^*$ can mis-prune

Why does **LCFS** need no extra condition, but  $A^*$  does?

**LCFS** — sorts by  $g(n)$ :

- first pop of  $n$  already has the smallest  $g(n)$  — safe.

$A^*$  — sorts by  $f = g + h$ :

- $h(n)$  cancels at  $n$ , but differs across prefixes  $p, p'$ ;
- a big  $h$ -gap can pop the larger- $g$  path first  $\rightarrow n$  committed to a worse prefix.

# How consistency restores the guarantee

Bound  $h$ -change along every edge  $p \rightarrow n$ :

$$h(p) - h(n) \leq c(p, n) \Leftrightarrow h(p) \leq c(p, n) + h(n)$$

— that's exactly **consistency**.

## Chain of consequences:

- $f$  is non-decreasing along every path
- first pop of  $n$  has the smallest  $g$  — LCFS guarantee, restored
- $\therefore$  multi-path pruning is safe

# Five search algorithms

| Algorithm | Frontier key  | Space    | Time     | Complete?        | Optimal?                             |
|-----------|---------------|----------|----------|------------------|--------------------------------------|
| DFS       | LIFO (stack)  | $O(bm)$  | $O(b^m)$ | No <i>cycles</i> | No                                   |
| BFS       | FIFO (queue)  | $O(b^d)$ | $O(b^d)$ | Yes              | Yes <i>unit cost</i>                 |
| LCFS      | $g(n)$        | $O(b^d)$ | $O(b^d)$ | Yes              | Yes                                  |
| GBFS      | $h(n)$        | $O(b^m)$ | $O(b^m)$ | No               | No                                   |
| A*        | $g(n) + h(n)$ | $O(b^m)$ | $O(b^m)$ | Yes              | Yes <i>admissible <math>h</math></i> |



## Learning goals (recap)

- ✓ Motivate heuristic search
- ✓ Trace LCFS, GBFS, A\*
- ✓ Analyze space, time, completeness, optimality
- ✓ Design and dominate admissible heuristics
- ✓ Prune the search space (cycle, multi-path, consistency)